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PROFESSIONAL EXPERIENCE

2025 – Present, Research Associate, Woods Hole Oceanographic Institution, Woods Hole, MA, Department of Marine Chemistry and Geochemistry

2022 – 2025, Postdoctoral Investigator, Woods Hole Oceanographic Institution, Woods Hole, MA, Department of Marine Chemistry and Geochemistry

EDUCATION

University of Tennessee	Knoxville, US
Ph.D. Environmental Engineering	Aug. 2022

University of Chinese Academy of Sciences	Beijing, China
M.S. Environmental Science	Jul. 2017

University of Copenhagen	Copenhagen, Denmark
M.S. Water and Environment	Mar. 2017

University of Jinan	Jinan, China
B.Eng. Environmental Engineering	Jun. 2014

RESEARCH INTERESTS

- Contaminant fate and transport
- Physical-chemical processes
- Water quality
- Environmental health
- Greenhouse gases
- Environmental sustainability
- Science communication

PUBLICATIONS

Peer-reviewed Papers:

1. Liu, S., Li, Y., Zeng, X., **Sun, Y.**, Huang, Z., Du, C., Lang, J., Wang, S., and Jia, Y. 2026. Reversible control of microbial As(III) oxidation by nitrous oxide availability in flooded paddy soils. *Environmental Science & Technology*. In press.
2. Zhou, J., Sang, C., Liu, Z., Li, W., **Sun, Y.**, Zhang, J., Wang, S., and Zeng, X. 2026. Bioconversion of selenite to selenium nanoparticles by *Paenibacillus polymyxa* P10: metabolic basis and potential for wastewater treatment. *Applied and Environmental Microbiology*. In press.

3. **Sun, Y.**, Wu, X., Zanina, O. G., Rivkina, E. M., Lloyd, K. G., Löffler, F. E., and Vishnivetskaya, T. A. 2026. Incomplete denitrifying bacteria drive N₂O fluxes in ancient Siberian permafrost microcosms. *FEMS Microbiology Ecology*. 102(5), fiag034.
4. **#Sun, Y.**, James, B. D., Houston, R. K., Edwards, B., Izallalen, M., Mazumde, S., Shankar R., Reddy, C. M., and Ward, C. P. 2025. Temperature controls the lifetimes and microbial communities degrading cellulose diacetate in the coastal ocean. *Environmental Science & Technology*. 59, 36, 19468–19478.
5. James, B. D., **Sun, Y.**, Pate, K., Irving, B., Ward, C. P. 2025. Designing circular, sustainable, and non-persistent consumer plastic products: a case study of drinking straws. *Environmental Science & Technology*. 59, 34, 18177–18189.
6. Murdoch, F. K., **Sun, Y.**, Fuller, M. E., Lilly, J., Wilson, J., Mullins, L., Hill, A., Löffler, F. E., and Kucharzyk, H. K. 2025. Presence of micro and nanoplastics affects degradation of chlorinated solvents. *Toxics*, 13(8), 656.
7. Waldrop, M. P., Ernakovich, J. G., Schaefer, S. R., Mackelprang, R., Tas, N., Vishnivetskaya, T. A., Li, Z., Ricketts, M. P., Leewis, M. C., Hewitt, R., Hultman, J., **Sun, Y.**, et al. 2025. Microbial ecology of permafrost soils: populations, processes, and perspectives. *Permafrost and Periglacial Processes*, 0:1–14.
8. **#Sun, Y.**, Yin, Y., He, G., Cha, G., Ayala-del-Río, H., Konstantinidis, K. T., and Löffler, F. E. 2024. pH selects for distinct N₂O-reducing microbiomes in tropical soil microcosms. *ISME Communications*, ycae070.
9. James, B. D., **Sun, Y.**, Pate, K., Perri, S. T., Shankar, M., Reddy, C. M., and Ward, C. P. 2024. Foaming enables material-efficient bioplastic products with minimal persistence. *ACS Sustainable Chemistry & Engineering*, 12, 43, 16030–16040.
10. Sun, H., Radosevich, M., **Sun, Y.**, Millet, L., Qian, S., and Zhuang, J. 2024. Pore confinement enhances but surface adhesion reduces bacterial cell-to-cell interaction. *Biology and Fertility of Soils*, 1-10.
11. Yin, Y., Murdoch F. K., Murdoch R., Yan J., Chen G., Xie Y., **Sun Y.**, and Löffler, F. E. 2024. An underappreciated greenhouse gas emission feedback: Nitrous oxide inhibition of methanogenesis. *The ISME Journal*, p.wrae027.
12. James, B. D., **Sun, Y.**, Perri, S. T., Edwards, B., Reddy, C. M., and Ward, C. P. 2024. Strategies to reduce the environmental lifetimes of drinking straws in the coastal ocean. *ACS Sustainable Chemistry & Engineering*, 12, 6, 2404–2411.
13. Yang, S., Shobnam, N., **Sun, Y.**, Löffler, F. E., and Im, J. 2024. The relative contributions of Mn(III) and Mn(IV) in manganese dioxide polymorphs to bisphenol A degradation. *Journal of Hazardous Materials*, 461, 132596.
14. **Sun, Y.**, Mazzotta, M. G., Miller, C. A., Apprill, A., Izallalen, M., Mazumde, S., Perri, S. T., Edwards, B., Reddy, C. M., and Ward, C. P. 2023. Distinct microbial communities degrade cellulose diacetate bioplastics in the coastal ocean. *Applied and Environmental Microbiology*, e01651-23.
15. Zhang, L., Yin, Y., **Sun, Y.**, Liang, X., Graham, D., Pierce, E; Löffler, F. E., and Gu, B. 2023. Inhibition of methylmercury and methane formation by nitrous oxide in Arctic tundra soil microcosms. *Environmental Science & Technology*, 57(14), 5655-5665.

16. **Sun, Y.**, Wang, C., May, A. L., Chen, G., Yin, Y., Im, J., Campagna, S., and Löffler, F. E. 2023. Mn(III)-mediated bisphenol A degradation: Mechanisms and products. *Water Research*, 235, 119787.
17. Xie, Y., Ramirez, D., Chen, G., He, G., **Sun, Y.**, Murdoch, F. K., and Löffler, F. E. 2023. Genome-wide expression analysis unravels fluoroalkane metabolism in *Pseudomonas* sp. strain 273. *Environmental Science & Technology*, 57(42), 15925–15935
18. Sun, H., **Sun, Y.**, Jin, M., Ripp, S. A., Sayler, G. S., and Zhuang, J. 2022. Domestic plant food loss and waste in the United States: Environmental footprints and mitigation strategies. *Waste Management*, 150, 202-207.
19. **Sun, Y.**, Im, J., Shobnam, N., Fanourakis, S. K., He, L., Anovitz, L. M., Erickson R. P., Sun, H., Zhuang, J., and Löffler, F. E. 2021. Degradation of adsorbed bisphenol A by soluble Mn(III). *Environmental Science & Technology*, 55(19), 13014-13023.
20. Shobnam, N., **Sun, Y.**, Mahmood, M., Löffler, F. E., and Im, J. 2021. Biologically mediated abiotic degradation (BMAD) of bisphenol A by manganese-oxidizing bacteria. *Journal of Hazardous Materials*, 417, 125987.
21. **Sun, Y.**, Zeng, X., Yang, L., Shi, Y., Chen, X., and Zhuang, J. 2017. Effects of strong reductive process on transformation of heavy metals in protected vegetable soil. *The Journal of Applied Ecology*, 28(11), 3759-3766.

Manuscript in Review/in Preparation:

22. **Sun, Y.**, Grim, S. L., James, B. D., Reddy, C. M., and Ward, C. P. Photodegradation of plastics shapes the diversity of microbial communities and functions in seawater. [ChemRxiv](#). In revision at *Environmental Science & Technology*.
23. Ward, C. P., Schwartz, G., **Sun, Y.**, Messenger, S., Frye-Jones, J., Italiane, D., Swarr, G., Aguilera, M., Rodgers, R. Chronic sunlight-driven mobilization of persistent, bioactive carbon and vanadium from asphalt to downstream aquatic systems. In review at *PNAS*.
24. **Sun, Y.**, Ahern, O., Huber, J., Reddy, C. M., and Ward, C. P. Stable-isotope probing of microbial species driving cellulose diacetate biodegradation in the coastal ocean. In preparation.
25. **Sun, Y.**, Reddy, C. M., and Ward, C. P. A next-generation method for quantifying plastic biodegradation in seawater. In preparation.
26. **Sun, Y.**, Shuai, D., Reddy, C. M., and Ward, C. P. Boosting biodegradation of cellulose diacetate bioplastic by graphitic carbon nitride and sunlight in seawater. In preparation.

Conference Papers:

1. **Sun, Y.**, Im, J., He, L., Erickson, P., and Löffler, F. E. 2022. Is adsorbed bisphenol A (BPA) degradable? *Eleventh International Conference on the Remediation and Management of Contaminated Sediments*. Best paper award.
2. **Sun, Y.**, Im, J., He, L., Erickson R. P., and Löffler, F. E. 2022. Soluble Mn(III)-mediates degradation of adsorbed bisphenol A. *Thirteenth Annual Geosyntec Student Paper Competition*. Third-place award.

RESEARCH GRANT EXPERIENCE

In Preparation:

1. Assessing the impacts of plastic-derived organic carbon on aquatic microbiomes across freshwater to ocean continuum. **PI**. To be submitted to NSF Environmental Engineering Program.
2. Collaborative Research: How subduction-derived volatiles support the deep biosphere in the warm cascadia forearc. **Co-PI**. To be submitted to NSF Chemical Evolution of the Solid Earth and Volcanology Program.

Awarded:

3. Characterizing the distribution of bisphenol A in pores of microscale activated carbon. The Neutron Sciences General User Proposal (IPTS#: 24752.1), Oak Ridge National Laboratory, 2020. **PI**.

Application Experience:

4. Transformation of PFAS in AFFF-contaminated soils: Insights from a pilot greenhouse study to mimic contaminated sites. DOD-SERDP. 2025. **Co-PI**. Not funded.
5. Poly-and perfluoroalkyl substances (PFASs) occurrence and microbial community responses in coastal wetland ecosystem. Northeastern University and Woods Hole Oceanographic Institution Seed Grant Program, \$100k. 2023. **Co-PI**. Not funded.
6. Stable isotope probing of the microbial species and enzymes that drive the biodegradation of cellulose diacetate bioplastics in the ocean. Simons Postdoctoral Fellowships in Marine Microbial Ecology, Simons Foundation, \$270k. 2023. **PI**. Not funded.
7. Enhanced bioremediation of groundwater contaminants through sorption and biologically mediated abiotic degradation of inhibitory carbon tetrachloride. NIEHS Bioremediation Superfund, 2020. Major contributor. Not funded.

RESEARCH PERMITS AND FIELD APPLICATIONS

1. Yellowstone National Park Research Permit Application #8438: Assessing plastic degradation potential and plastic-associated microbial communities in acidic hydrothermal waters of Yellowstone National Park. National Park Service, Yellowstone Center for Resources. **Status: Under review; additional materials requested.**

HONORS AND AWARDS

- American Society for Microbiology Future Leaders Mentorship Fellowship, 2024
- Extraordinary Professional Promise Award, University of Tennessee, 2022
- Paper Competition Award, Geosyntec Consultants, Inc., 2022
- Paper Competition Award, Battelle Sediments Conference, 2022
- Graduate Student Senate Academic Support, University of Tennessee, 2022
- Graduate Student Senate Travel Award, University of Tennessee, 2019-2022
- Merit student, University of Chinese Academy of Sciences, 2016
- Outstanding Student Scholarship, University of Jinan, 2010-2013

TEACHING EXPERIENCE

Participation in Education Program:

- ASM Future Leaders Mentorship Fellowship Program, 2024-2026
- WHOI Summer Student Fellowship Program, May-Aug 2024, 2025

- Geosciences Education & Mentorship Support Program, 2023-2024
- *Inclusive Teaching Certificate*, University of Tennessee, 2023
- *Course-Based Assessment Certificate*, University of Tennessee, 2023
- Student Mentoring and Research Training Program, University of Tennessee–Oak Ridge Innovation Institute, May-Aug 2022
- Summer School for High School Students, China, Jul-Aug 2014

Mentoring:

- Kyle DeCruccio, undergraduate student, Cape Cod Community College, Jun-Aug, 2025
- Mallory Kastner, Ph.D. student, MIT-WHOI Joint Program, Feb-Mar, 2025
- Dominic Italiane, undergraduate student, University of Massachusetts-Boston, May, 2024-now
- Shinhyeong Choe, Ph.D. student, Woods Hole Oceanographic Institution, Feb-Mar, 2024
- Kimberly Lutchman, undergraduate student, Queens College, CUNY, Aug 2023-2024
- Devan Linson, undergraduate student, Washington State University, May-Aug, 2022
- Divinity Oliver, undergraduate student, Washington State University, May-Aug 2022
- Jessica Rattray, undergraduate student, Vassar College, May-Aug, 2022

PROFESSIONAL DEVELOPMENT AND SERVICES

Professional Development:

- Proposal Writing Workshop, Woods Hole Oceanographic Institution, 2023
- Academic Job Application Workshop, Woods Hole Oceanographic Institution, 2023

Service:

- Poster judge, 2025 AEESP Conference.
- Topic reviewer editor for *Frontiers in Microbiology* and *Frontiers in Marine Science*.
- Reviewer for *Biodegradation*, *Bioresource Technology*, *Chemosphere*, *Ecological Processes*, *Ecotoxicology*, *Environmental Sciences Europe*, *Environmental Geochemistry and Health*, *Environmental Monitoring and Assessment*, *Environmental Research*, *Environmental Science & Technology*, *Frontiers in Microbiology*, *Frontiers in Marine Science Journal of Geophysical Research – Biogeosciences*, *Journal of Hazardous Materials Advances*, *Journal of Hazardous Materials Letters*, *Reviews of Environmental Contamination and Toxicology*, and *Water Research*.

Outreach:

- Massachusetts Region V Science and Engineering Fair, 2024
- Reverse Science Fair, Falmouth High School, MA, 2023

Professional Affiliation:

- American Chemical Society
- American Geophysical Union
- American Society for Microbiology
- Association of Environmental Engineering and Science Professors

SELECTED PRESENTATIONS (presenter underlined)

Oral:

1. Sun, Y., Ahern, O., Huber, J., Reddy, C. M., and Ward, C. P. 2026. Stable-isotope probing of microbial species driving cellulose diacetate biodegradation in the coastal ocean. American Chemical Society Spring Meeting, Atlanta, GA.
2. Sun, Y. Environmental pollution and health: insights from plastic degradation. 2025. Morgan State University, Baltimore, MD. *Invited*.
3. Sun, Y. Unravelling plastic degradation in the coastal ocean. 2025. Woods Hole Oceanographic Institution, Woods Hole, MA. *Invited*.
4. Sun, Y., Frye-Jones, J., Grim, S., James, B., Reddy, C. M., and Ward, C. P. 2025. Photodegradation of plastics shapes the diversity of microbial communities and functions in seawater. AEESP Conference, Durham, NC.
5. Sun, Y., James, B., Reddy, C. M., and Ward, C. P. 2025. Temperature controls the timelines and microbial communities degrading cellulose diacetate in the coastal ocean. American Chemical Society Spring Meeting, San Diego, CA.
6. Sun, Y., Grim, S., James, B., Reddy, C. M., and Ward, C. P. 2025. Photodegradation of plastics shapes the diversity of microbial communities and functions in seawater. American Chemical Society Spring Meeting, San Diego, CA.
7. Sun, Y. Potential of microbial degradation of plastics in the ocean. 2025. AAAS Annual Meeting, Boston, MA.
8. Sun, Y. Understanding the fate of conventional and next-generation plastics in the ocean. 2024. University of Massachusetts, Dartmouth, MA. *Invited*.
9. Sun, Y., Mazzotta, M. G., Miller, C. A., Apprill, A., Izallalen, M., Mazumde, S., Perri, S. T., Edwards, B., Reddy, C. M., and Ward, C. P. 2024. Unique microbes degrade cellulose diacetate bioplastics in the ocean. American Chemical Society Spring Meeting, New Orleans, LA.
10. Sun, Y., Ward, C. P., James, B., and Reddy, C. M. 2023. Translating fundamental knowledge about the degradation of cellulose diacetate in the environment into high-utility, low-persistence consumer products. Eastman Chemical Company, Kingsport, TN. *Invited*.
11. Sun, Y. 2023. "Bioplastics meet biogeochemical cycles", WHOI's Microplastics Initiative. Woods Hole Oceanographic Institution, Woods Hole, MA.
12. Sun, Y., Mazzotta, M. G., Miller, C. A., Apprill, A., Reddy, C. M., and Ward, C. P. 2023. Distinct microbial communities degrade cellulose diacetate plastics in the coastal ocean. AEESP Conference, Boston, MA.
13. Sun, Y. 2022. "Role of dissolved Mn(III) in the transformation of adsorbed bisphenol A", webinar presentation. Geosyntec Consultants, Inc. *Invited*.
14. Sun, Y., Im, J., He, L., Erickson, P., and Löffler, F. E. 2022. Role of dissolved Mn(III) in the transformation of adsorbed bisphenol A. Eleventh International Conference on the Remediation and Management of Contaminated Sediments, Nashville, TN. *Invited*.

Poster:

15. Sun, Y., James, B., Ahern, O., Huber, J., Reddy, C. M. and Ward, C. P. 2026. A systematic evaluation of consumer plastics degradation in coastal environments. Gordon Research Conference, Environmental Sciences: Water. Holderness, NH.

16. **Sun, Y.**, Wu, X., Zanina, O., Rivkina, E., Lloyd, K., Löffler, F. E., and Vishnivetskaya, T. 2022. Nitrous oxide emissions from thawing permafrost: A microcosm study. American Geophysical Union Fall Meeting, Chicago, IL.
17. **Sun, Y.**, Yin, Y., Chen, G., Cha, G., Ayala-del-Río, H., Konstantinidis, K. T., and Löffler, F. E. 2022. pH selects for distinct N₂O-reducing microbiomes in tropical soil microcosms. Eighteenth International Symposium on Microbial Ecology, Lausanne, Switzerland.
18. **Sun, Y.**, Im, J., He, L., Erickson, P., and Löffler, F. E. 2022. Role of dissolved Mn(III) in the transformation of adsorbed bisphenol A. Eleventh International Conference on the Remediation and Management of Contaminated Sediments, Nashville, TN.
19. **Sun, Y.**, Shobnam, N., Im, J., Chen, G., Yin, Y., Erickson, P. R., and Löffler, F. E. 2019. Degradation of adsorbed BPA by oxidized manganese species. American Geophysical Union Fall Meeting, San Francisco, CA.
20. **Sun, Y.**, Shobnam, N., Im, J., Chen, G., Yin, Y., Campagna, S., and Löffler, F. E. 2019. Biologically mediated abiotic degradation of bisphenol A. American Chemical Society Spring Meeting, Orlando, FL.